DESIGN PROJECT · HALLUCINATION DETECTION IN LLMS

LLM Hallucination Detection using Multi-Layer EigenScore

Detecting confident-but-wrong answers from a language model's internal states

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Builds on Chen et al., INSIDE, ICLR 2024

01 The problem

Large language models **hallucinate**: they give fluent, confident answers that are factually wrong, in exactly the same tone as correct ones. Detection must work **without the right answer** and without external fact-checking.

Q: In which year did the bridge open?

Model knows

1932 1932 1932 1932

answers agree → trust

Model guesses

1932 1936 1928 1941

answers scatter → likely hallucination

02 Background: INSIDE & EigenScore

INSIDE measures how much several sampled answers **disagree in meaning**, using the model's hidden states instead of its output words.

Sample K = 10 answers → **Embed** middle layer, last token → **Covariance** 10 × 10 → **Eigenvalues** spread → **Score** high = risky

$$\text{EigenScore} = (1/K) \cdot \log\det(\Sigma + \alpha I) = (1/K)\,\Sigma_i \log \lambda_i$$

Consistent answers → eigenvalues collapse → low score. Diverse answers → large eigenvalues → high score. **Test-time feature clipping** additionally truncates extreme activations in the penultimate layer (0.2% / 99.8% percentiles, memory bank of 3,000 tokens) to expose over-confident errors.

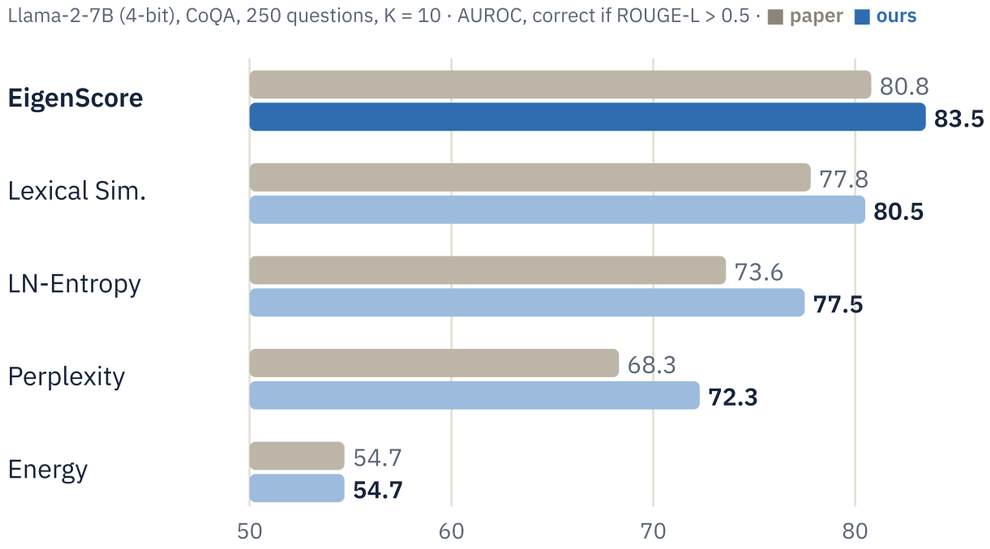
03 Our implementation

12	1	8 GB	8/8
blocking bugs fixed in the official code	missing component implemented: feature clipping	GPU runs a 7B model (4-bit, middle-layer hook)	automated tests passing

- Fixed a batching bug that scored only **2 of the 10** answers.
- **FeatureClipper:** ring-buffer memory bank, per-neuron percentile thresholds, warm-up, and a forward hook on the penultimate layer so clipping acts during generation.
- Memory-efficient capture of a single layer, verified identical to the full path.

04 Reproduction results

Llama-2-7B (4-bit), CoQA, 250 questions, K = 10 · AUROC, correct if ROUGE-L > 0.5 · ■ paper ■ ours

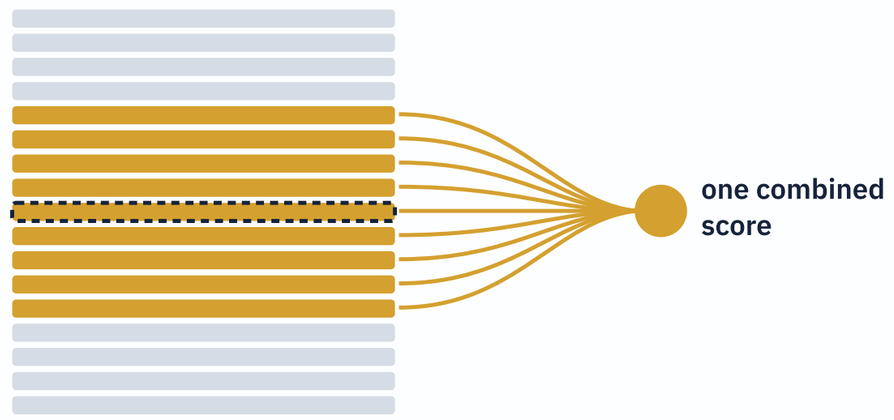


The paper's ranking reproduces exactly. EigenScore beats the strongest baseline by **+2.9 AUROC** (95% CI [−0.6, +6.5], paired bootstrap). The feature-clipping gain is **inconclusive** at this scale: −1.0, CI [−4.7, +2.6], against the paper's +1.1.

05 Limitations we identified

- **One hand-picked layer.** EigenScore reads only layer L/2, yet the paper's own ablation shows AUROC varies by several points across layers. The best layer is model-specific.
- **Scale-dominated covariance.** A few very large ("massive") activation dimensions can dominate the covariance instead of semantic differences.
- **Blind to consistent errors.** When the model repeats the same wrong answer, all samples agree and the score trusts it; clipping helps only marginally.

06 Proposed method: Multi-Layer EigenScore



dashed: original single layer (L/2)
gold: band of middle layers, L/4 … 3L/4

1. **Multi-layer aggregation.** Average EigenScore over a fixed band of middle layers (L/4 to 3L/4; layers 8–24 of 32). This removes the hand-picked layer and averages out per-layer noise, like averaging several thermometers.
2. **Scale normalisation.** L2-normalise each answer embedding (or standardise each dimension using a memory bank) before the covariance, so the score reflects differences in meaning, not activation magnitude.
3. **Confidence-aware hybrid.** Rank-average Multi-Layer EigenScore with token-level confidence (LN-Entropy). When all answers agree, token confidence still signals uncertainty, targeting consistent hallucinations.

$$\text{ML-ES} = \text{mean}_{l \in [L/4,\ 3L/4]}\ \text{EigenScore}_l$$

All variants are training-free and computed from the same K samples: no extra generation, only more layers read.

07 Evaluation plan in progress

Item	Setting
Model	Llama-2-7B, 4-bit, RTX 4060 (8 GB)
Datasets	CoQA · TriviaQA · NQ-Open, 1,000 questions each
Sampling	K = 10, temperature 0.5, top-p 0.99
Metric	AUROC (ROUGE-L > 0.5 correctness)
Significance	paired bootstrap, 10,000 resamples
Primary test	**Multi-Layer EigenScore vs. original EigenScore**

Layer band and settings are fixed in advance to avoid tuning on test data. Extra analyses: per-layer AUROC curve, and detection rate on consistently-wrong answers.

08 Conclusion

We restored a working implementation of INSIDE, completed its missing feature-clipping component, and reproduced its central finding on consumer hardware: **internal-state consistency (EigenScore) is the strongest hallucination signal** among the methods compared. Our extension targets its main weaknesses, the single-layer choice and the blind spot for consistent errors, with a training-free **Multi-Layer EigenScore** and a confidence-aware hybrid.

Reference: C. Chen et al., "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection," ICLR 2024. Code: github.com/alibaba/eigenscore